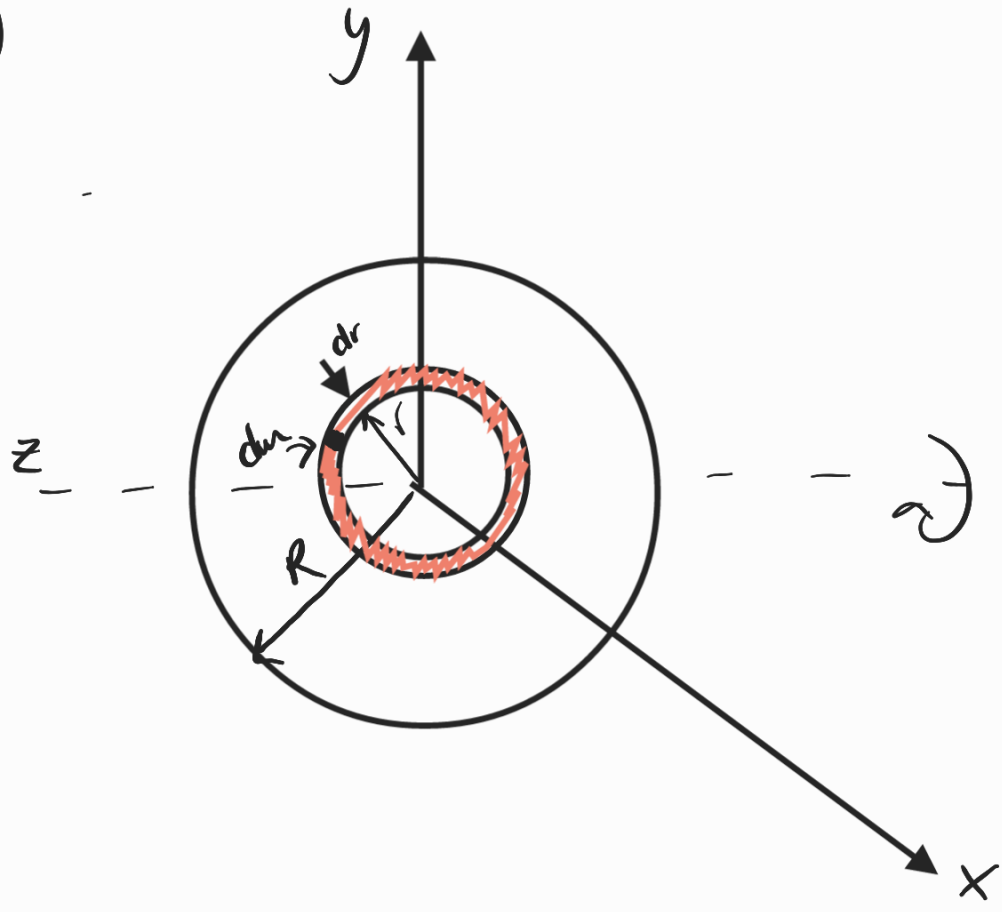
(11)



Εφόσον ο δίσκος είναι λεπτός και ομογενής, υποθέτουμε ότι η μάζα του είναι ομοιόμορφα κατανεμημένη.

$$I = \int r^2 \, dm$$

$$dm = \frac{dA}{A} \cdot M \quad (1) \qquad \text{όπου} \quad A = \pi R^2 \quad (2a)$$

$$\Rightarrow \frac{dA}{dr} = 2\pi r \Rightarrow dA = 2\pi r \, dr \quad (2b)$$

(2) → (1): $dm = \frac{2\pi r \, dr}{\pi R^2} \cdot M$

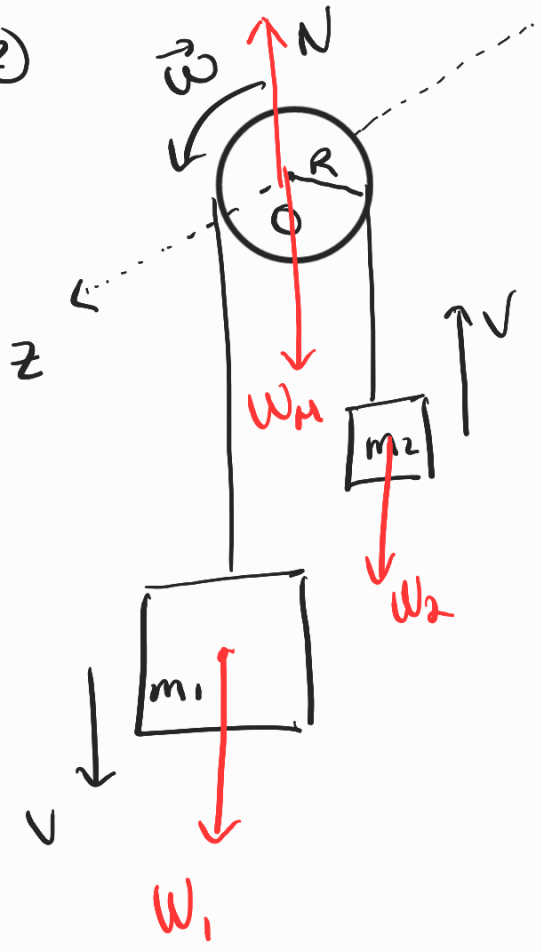
$$\Rightarrow I = \int_0^R \frac{r^2 \cdot 2\pi r M}{\pi R^2} dr$$

$$= \frac{2M}{R^2} \int_0^R r^3 dr$$

$$= \frac{2M}{R^2} \left[\frac{r^4}{4} \right]_0^R = \frac{2M}{R^2} \frac{R^4}{4_2}$$

$$\Rightarrow \boxed{I = \frac{1}{2} MR^2}$$

(2)



$a_r = ?$

$a = ?$

Σύστημα: ότι κινείται δηλ μάζες, τροχαλία

$$I = \frac{1}{2} M R^2$$

$$\sum \tau_z = \frac{dL_z}{dt} = \tau_N + \tau_w + \tau_{w_1} + \tau_{w_2}$$

$$= N(0) + W(0) + W_1 R + W_2 R$$

$$= 0 + 0 + m_1 g R - m_2 g R$$

$$= (m_1 - m_2) g R$$

(περιστροφική ↑ γύρω από ΚΜ, περιστροφικές ↑ γύρω από άξονα)

$$\sum L_z = L_\tau + L_{m_1} + L_{m_2} = I\omega + m_1 v R + m_2 v R$$

$$m_1 g R - m_2 g R = \frac{d}{dt}\left[m_1 v R + m_2 v R + I\omega\right]$$

$$\Rightarrow m_1 g R - m_2 g R = m_1 \frac{dv}{dt} R + m_2 \frac{dv}{dt} R + I \frac{d\omega}{dt}$$

$$\Rightarrow m_1 g R - m_2 g R = m_1 a R + m_2 a R + I \alpha_r$$

και $a = a_r \cdot r \Rightarrow a_r = \frac{a}{R}$

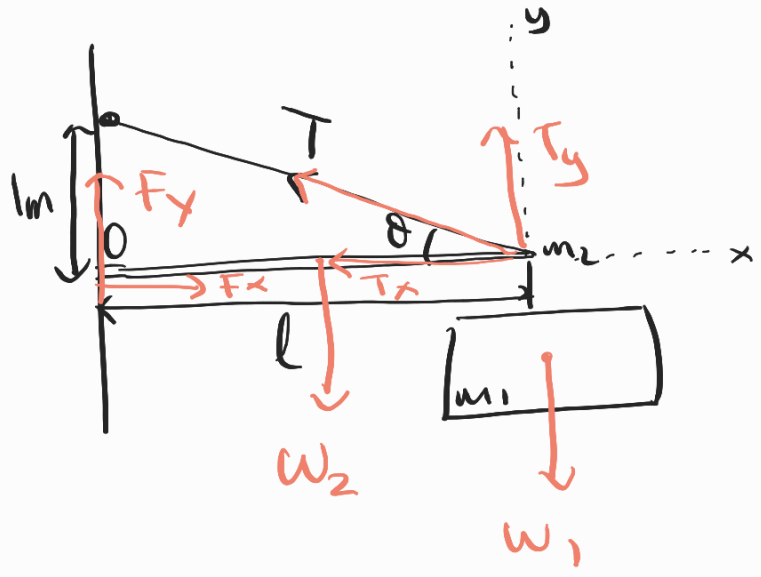
$$\Rightarrow (m_1 - m_2) g R = (m_1 + m_2) a R + \frac{M R^2 a}{2R}$$

$$\Rightarrow a\left(m_1 + m_2 + M/2\right) = (m_1 - m_2) g$$

$$\Rightarrow a = \frac{(m_1 - m_2) g}{m_1 + m_2 + M/2} \quad //$$

$$a_r = \frac{a}{R} = \frac{m_1 - m_2}{m_1 + m_2 + M/2} \cdot \frac{g}{R} \quad //$$

③



$l = 2m$

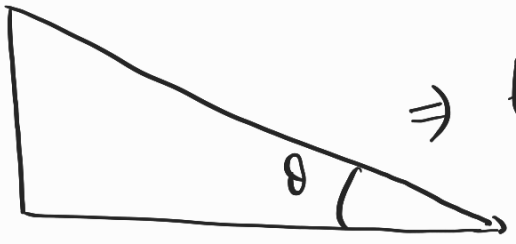
$m_1 = 20kg$

$m_2 = 4\ kg$

$\vec{T} = ?$

$\vec{F} = ?$

1



$\Rightarrow \tan\theta = \frac{1}{2} \Rightarrow \theta = 26.6°$

2

$$\Sigma F_x = T_x - F_x = 0 \quad ①$$

$$\Sigma F_y = F_y + T_y - w_2 - w_1 = 0 \quad ②$$

$$T_x = T\cos\theta$$

$$T_y = T\sin\theta$$

$$\Rightarrow \boxed{F_x = T\cos\theta} \quad ③$$

$$\Rightarrow \boxed{F_y + T\sin\theta = (m_2 + m_1)g} \quad ④$$

Powes:

$$\Sigma \tau = \tau_{T_y} + \tau_{F_y} - \tau_{w_1} - \tau_{w_2} = 0$$

$$\Rightarrow T_y \cdot l + F_y \cdot 0 - m_1 g l - m_2 g \cdot l/2 = 0$$

$$\Rightarrow T \sin\theta \cdot l = m_1 g l + m_2 g l/2$$

$$\Rightarrow 2T\sin\theta = (20)(10)(2) + (4)(10)(1)$$

$$\Rightarrow T = \frac{400+40}{2\sin(26.6)} = 491\ N$$

$$\Rightarrow ③: F_x = 491\cos(26.6) = 439\ N \quad N$$

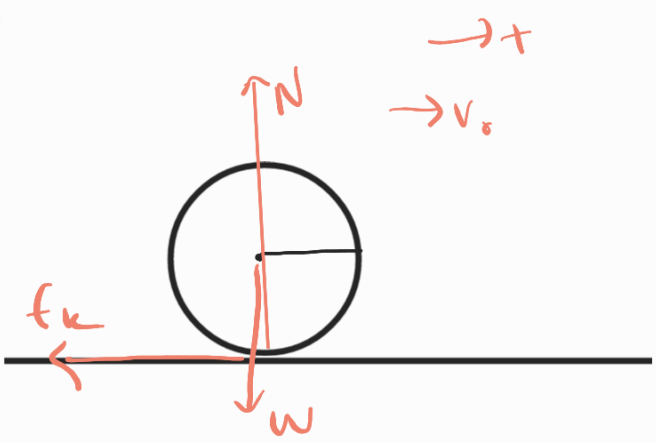
$$④ \quad F_y = (20+40)10 - 491 \cdot \sin(26.6)$$

$$\Rightarrow F_y = 240 - (491)(0.45)$$

$$= 20\ N$$

$$\Rightarrow \vec{F} \cong 439\ \hat{i} + 20\ \hat{j}\ N$$

④



$m = 0.16\,kg$

$r = 0.03\,m$

$v_0 = 4\,m/s$

$\mu_k = 0.6$

(a) $\sum F_y = 0 \Rightarrow N = W = m \cdot g$

$\sum F_x = -f_k = ma$

$\Rightarrow -\mu_k N = m \cdot a$

$\Rightarrow -\mu_k m \cdot g = ma$

$\Rightarrow a_{cm} = -\mu_k g$ ① ⇒ a gravitation too KM

$\sum \tau = I_{cm} a_r \rightarrow$ a rotation

$\Rightarrow f_k \cdot r + W \cdot 0 + N \cdot 0 = I a_r$

$\Rightarrow \mu_k \cdot m \cdot g \cdot r = I a_r$

$I_{sphere} = \frac{2}{5} m r^2$

$\Rightarrow \mu_k \cdot m \cdot g \cdot r = \frac{2}{5} m r^2 a_r$

$\Rightarrow a_r = \frac{5 \mu_k \cdot g}{2r}$ ②

$$v_{cm} = v_{0cm} + at \quad (3)$$

$(1) \rightarrow (3)$: $v_{cm} = v_0 - \mu_k g t$

$$\omega = \omega_0 + \alpha_r t \quad (4)$$

$(2) \rightarrow (4)$: $\omega = \frac{5}{2} \frac{\mu_k g t}{r}$

and $v_{cm} = \omega r$

$$\Rightarrow v_0 - \mu_k g t = \frac{5}{2} \frac{\mu_k g t}{r} \cdot r$$

$$\Rightarrow v_0 - \frac{7}{2} \mu_k g t = 0$$

$$\Rightarrow t = \frac{2v_0}{7\mu g} = \frac{2(4)}{7(0.6)(10)} = 0.19\ s$$

(b) $\Delta x = v_0 t + \frac{1}{2} a t^2$

$$= \frac{v_0 2v_0}{7\mu g} - \frac{\mu g}{2} \left(\frac{2v_0}{7\mu g} \right)^2$$

$$= \frac{12}{49} \frac{v_0^2}{\mu g} = \frac{12 (4)^2}{49 (0.6)(10)} = 0.67 m$$

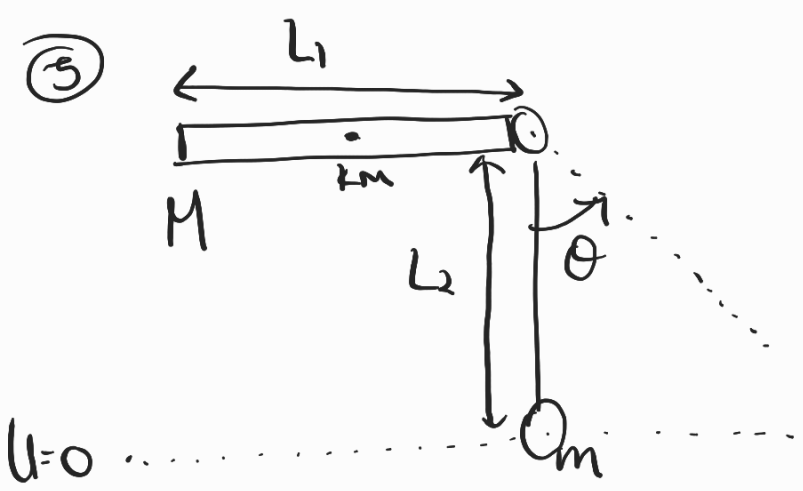
(γ) $V = V_0 - \mu_k g t$

$$\Rightarrow V = V_0 - \frac{\mu_k g \, 2V_0}{7\mu_k g}$$

$$\Rightarrow V = V_0 - \frac{2}{7}V_0 = \frac{5}{7}V_0$$

$$= \frac{5}{7}(4) = 2.9 \text{ m/s}$$

⑤



$M = 2\ kg$

$L_1 = 1.2\ m$

$L_2 = 0.8\ m$

$\vartheta = 37°$

(α) $m = ?$

Εφαρμόζω διατήρηση ΜΕ από αρχική θέση μέχρι κατώτατο σημείο.

$\Rightarrow E_i = M \cdot g \cdot L_1$

Ενέργεια στην αρχή $= E_f = K_{rot} + M g L_1/2$

$\Rightarrow E_i = E_f$

$\Rightarrow Mg L_1 = Mg \frac{L_1}{2} + \frac{1}{2} I \omega^2$

$I = \frac{1}{3} M L_1$

$\Rightarrow Mg L_1 = Mg L_1/2 + \frac{1}{2} \cdot \frac{1}{3} M L_1^2 \omega^2$

$\Rightarrow g - \frac{g}{2} = \frac{1}{6} L_1 \omega^2$

$\Rightarrow g/2 = \frac{L_1 \omega^2}{6} \Rightarrow \omega^2 = \frac{6g}{2L_1} = \frac{3g}{L_1}$

$\Rightarrow \omega = \sqrt{3g/L_1}$ ①

Μετά την κρούση η ω γίνεται ω'.
αλλά η στροφορμή διατηρείται

$$\Rightarrow \Delta L = L_f - L_i = 0$$

$$L_f = L_{ράβδω} + L_m = I_\rho \omega' + I_m \omega'$$

$$L_i = L_{ράβδω}$$

$$\Rightarrow \left(I_\rho + I_m\right)\omega' - I_\rho \omega = 0$$

$$\Rightarrow \omega' = \frac{I_\rho \cdot \omega}{I_\rho + I_m}$$

από ①

$$\Rightarrow \omega' = \frac{\frac{1}{3}ML_1^2}{\frac{1}{3}ML_1^2 + mL_2^2} \cdot \omega = \frac{\frac{1}{3}ML_1^2}{\frac{1}{3}ML_1^2 + mL_2^2} \cdot \sqrt{\frac{3g}{L_1}}$$

$$\Rightarrow \omega' = \frac{\left(\frac{1}{3}\right)(2)(1.2)^2}{\left(\frac{1}{3}\right)(2)(1.2)^2 + m(0.8)^2} \sqrt{\frac{(3)(10)}{(1.2)}}$$

$$= \frac{0.96}{0.96 + m(0.64)} \cdot 5$$

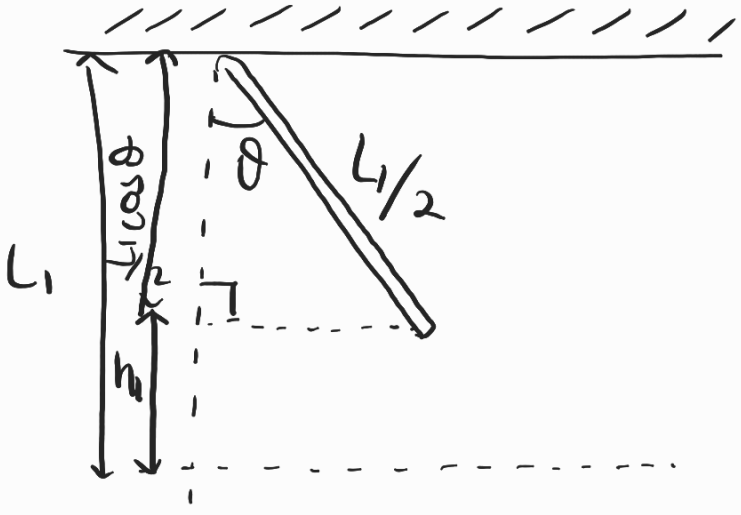
$$\Rightarrow \boxed{\omega' = \frac{4.8}{0.96 + 0.64m}} \quad ②$$

ΑΔΕ μετά την κρούση σε $\vartheta=0°$ (E_3) και σε

$\vartheta=37°$ (E_4) για ράβδο +m:

$$E_3 = \frac{1}{2} I \omega'^2 + Mg\frac{L_1}{2} + mg(L_1 - L_2)$$

$$E_4 = Mgh_1 + mgh_2$$



$$\Rightarrow h_1 = L_1 - \frac{L_1}{2}\cos\vartheta$$

$$h_2 = L_1 - L_2\cos\vartheta$$

$$\Rightarrow E_4 = MgL_1 - \frac{MgL_1\cos\vartheta}{2} + mgL_1 - mgL_2\cos\vartheta$$

$$\Rightarrow E_3 = E_4$$

$$\Rightarrow \frac{1}{2} I \omega'^2 + \frac{MgL_1}{2} + mgL_1 - mgL_2 = MgL_1 - \frac{MgL_1\cos\vartheta}{2} + mgL_1 - mgL_2\cos\vartheta$$

$$\Rightarrow \frac{1}{2} I \omega'^2 - \frac{1}{2} MgL_1 + \frac{1}{2} MgL_1\cos\vartheta - mgL_2 + mgL_2\cos\vartheta = 0$$

$$\Rightarrow \frac{1}{2} I \omega'^2 - Mg\frac{L_1}{2}(1-\cos\vartheta) - mgL_2(1-\cos\vartheta) = 0$$

όπου $I = I_p + I_m = \frac{1}{3}ML_1^2 + mL_2^2$

$L_1 = 1.2$, $L_2 = 0.8$ και $\vartheta = 37$, $M = 2$

$\Rightarrow \frac{1}{2}I\omega'^2 - \frac{2}{2}\cdot 10\cdot 1.2(1-\cos 37) - 10m(0.8)(1-\cos 37) = 0$ (=0.2, =0.2)

$\Rightarrow \frac{1}{2}I\omega'^2 - 12(0.2) - 8(0.2)m = 0$

$\Rightarrow \frac{1}{2}I\omega'^2 - 2.4 - 1.6m = 0$

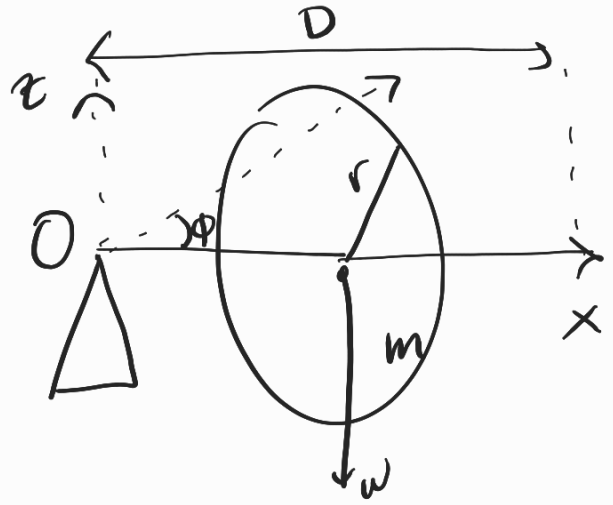
$\Rightarrow \frac{1}{2}\left(\frac{1}{3}ML_1^2 + mL_2^2\right)\frac{4.8^2}{(0.96+0.64m)^2} - 1.6m = 2.4$

$\therefore \boxed{m = 1.2\ \text{kg}}$

(b) $\Delta E = E_3 - E_0$

$=$

⑥



$D = 0.5\,m$

$R = 0.28\,m$

$W = 30\,N$

$f = 12\,Hz$

(a) $L_{\text{rotas}} = I\omega = MR^2\omega = \frac{W}{g} R^2\omega$

$\omega = 2\pi f = 2\pi(12) = 24\pi \text{ rad/s}$

$\Rightarrow L = \frac{30}{9.81} \cdot (0.28)^2 (24\pi) = 18\,\hat{i}\ Js$

(b) $\omega_p = \frac{d\phi}{dt} = \frac{M \cdot g \cdot D/2}{L} = \frac{\frac{W}{g} \cdot g \cdot \frac{D}{2}}{L}$

$= \frac{W \cdot D}{2L} = \frac{(30)(0.5)}{(2)(18)}\hat{i}$

$= 0.4\,\hat{i} \text{ rad/s}$

(c) $T_p = ?$

$T_p = \frac{2\pi}{\omega_p} = \frac{2\pi}{0.4} \simeq 15\,s$

(d) $L_p = ? = I_{CM}\omega_p = M\left(\frac{D}{2}\right)^2\omega_p = 0.07g\,\hat{k}\ Js$